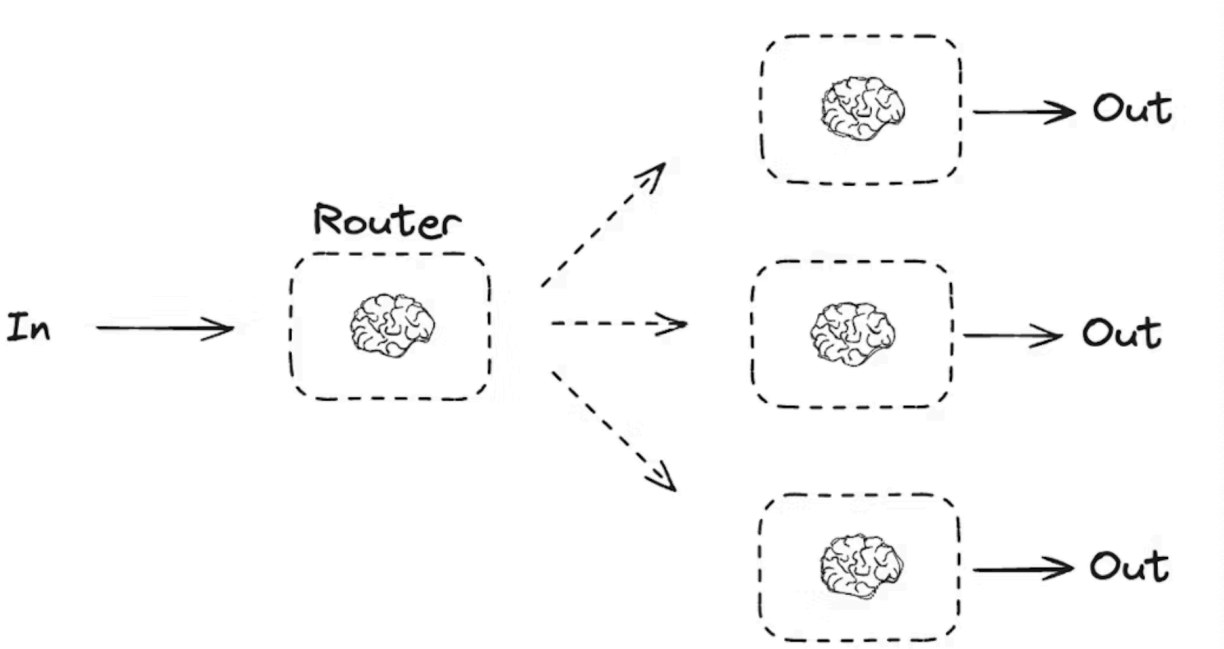
Routing




```
from typing_extensions import Literal
from langchain.messages import HumanMessage, SystemMessage


# Schema for structured output to use as routing logic
class Route(BaseModel):
    step: Literal["poem", "story", "joke"] = Field(
        None, description="The next step in the routing process"
    )


# Augment the LLM with schema for structured output
router = llm.with_structured_output(Route)
```

```
# State
class State(TypedDict):
    input: str
    decision: str
    output: str


# Nodes
def llm_call_1(state: State):
    """Write a story"""

    result = llm.invoke(state["input"])
    return {"output": result.content}


def llm_call_2(state: State):
    """Write a joke"""

    result = llm.invoke(state["input"])
    return {"output": result.content}


def llm_call_3(state: State):
    """Write a poem"""

    result = llm.invoke(state["input"])
    return {"output": result.content}


def llm_call_router(state: State):
    """Route the input to the appropriate node"""

    # Run the augmented LLM with structured output to serve as routing logic
    decision = router.invoke(
        [
            SystemMessage(
```

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            content="Route the input to story, joke, or poem based on the user's
request."
        ),
        HumanMessage(content=state["input"]),
    ]
  )

  return {"decision": decision.step}


# Conditional edge function to route to the appropriate node
def route_decision(state: State):
  # Return the node name you want to visit next
  if state["decision"] == "story":
    return "llm_call_1"
  elif state["decision"] == "joke":
    return "llm_call_2"
  elif state["decision"] == "poem":
    return "llm_call_3"


# Build workflow
router_builder = StateGraph(State)

# Add nodes
router_builder.add_node("llm_call_1", llm_call_1)
router_builder.add_node("llm_call_2", llm_call_2)
router_builder.add_node("llm_call_3", llm_call_3)
router_builder.add_node("llm_call_router", llm_call_router)

# Add edges to connect nodes
router_builder.add_edge(START, "llm_call_router")
router_builder.add_conditional_edges(
  "llm_call_router",
  route_decision,
  {  # Name returned by route_decision : Name of next node to visit
```

```
        "llm_call_1": "llm_call_1",
        "llm_call_2": "llm_call_2",
        "llm_call_3": "llm_call_3",
    },
)
router_builder.add_edge("llm_call_1", END)
router_builder.add_edge("llm_call_2", END)
router_builder.add_edge("llm_call_3", END)

# Compile workflow
router_workflow = router_builder.compile()

# Show the workflow
display(Image(router_workflow.get_graph().draw_mermaid_png()))

# Invoke
state = router_workflow.invoke({"input": "Write me a joke about cats"})
print(state["output"])
```